

रोल नं.

Roll No.



प्रश्न-पत्र कोड
Q.P. Code

55(B)/S

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

भौतिक विज्ञान (सैद्धान्तिक)
(केवल दृष्टिबाधित परीक्षार्थियों के लिए)

PHYSICS (Theory)

(FOR VISUALLY IMPAIRED CANDIDATES ONLY)

निर्धारित समय : 3 घण्टे

अधिकतम अंक : 70

Time allowed : 3 hours

Maximum Marks : 70

नोट	NOTE
(I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 27 हैं।	(I) Please check that this question paper contains 27 printed pages.
(II) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 33 प्रश्न हैं।	(II) Please check that this question paper contains 33 questions.
(III) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।	(III) Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
(IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।	(IV) Please write down the serial number of the question in the answer-book before attempting it.
(V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।	(V) 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

General Instructions :

Read the following instructions carefully and follow them :

- (i) *This question paper contains **33** questions. **All** questions are **compulsory**.*
- (ii) *This question paper is divided into **five** sections – **Sections A, B, C, D and E**.*
- (iii) *In **Section A** – Questions no. **1** to **16** are Multiple Choice type questions. Each question carries **1** mark.*
- (iv) *In **Section B** – Questions no. **17** to **21** are Very Short Answer type questions. Each question carries **2** marks.*
- (v) *In **Section C** – Questions no. **22** to **28** are Short Answer type questions. Each question carries **3** marks.*
- (vi) *In **Section D** – Questions no. **29** and **30** are case study based questions. Each question carries **4** marks.*
- (vii) *In **Section E** – Questions no. **31** to **33** are Long Answer type questions. Each question carries **5** marks.*
- (viii) *There is no overall choice given in the question paper. However, an internal choice has been provided in few questions in all the Sections except Section A.*
- (ix) *Use of calculator is **not** allowed.*

You may use the following values of physical constants wherever necessary :

$$c = 3 \times 10^8 \text{ m/s}$$

$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$\text{Mass of electron (} m_e \text{)} = 9.1 \times 10^{-31} \text{ kg}$$

$$\text{Mass of neutron} = 1.675 \times 10^{-27} \text{ kg}$$

$$\text{Mass of proton} = 1.673 \times 10^{-27} \text{ kg}$$

$$\text{Avogadro's number} = 6.023 \times 10^{23} \text{ per gram mole}$$

$$\text{Boltzmann constant} = 1.38 \times 10^{-23} \text{ JK}^{-1}$$

SECTION A

1. Three point charges $+q$, $-4q$ and $+2q$ are kept at the vertices of an equilateral triangle of side 'a'. The electrostatic potential energy of this system of charges is :

$$\begin{array}{ll} \text{(A)} & -\frac{1}{4\pi\epsilon_0} \left(\frac{q^2}{a^2} \right) \\ \text{(B)} & +\frac{1}{4\pi\epsilon_0} \left(\frac{8q^2}{a^2} \right) \\ \text{(C)} & +\frac{1}{4\pi\epsilon_0} \left(\frac{6q^2}{a} \right) \\ \text{(D)} & -\frac{1}{4\pi\epsilon_0} \left(\frac{10q^2}{a} \right) \end{array}$$

2. The element of a heater is cut into two equal parts. Now only one part is used in the heater. The heat generated now as compared to the original rating of the heater will be :

$$\begin{array}{ll} \text{(A)} & \text{halved} \\ \text{(B)} & \text{doubled} \\ \text{(C)} & \text{four times} \\ \text{(D)} & \text{same} \end{array}$$

3. A bar magnet of magnetic moment 4 JT^{-1} is kept along the direction of magnetic field of 0.4 T . The work done to rotate the magnet through an angle of 90° is :

$$\begin{array}{ll} \text{(A)} & \text{Zero} \\ \text{(B)} & -1.6 \text{ J} \\ \text{(C)} & -0.2 \text{ J} \\ \text{(D)} & 1.6 \text{ J} \end{array}$$

4. The dimensions of permeability of a medium are :

$$\begin{array}{ll} \text{(A)} & [\text{MLT}^{-1} \text{ A}^{-1}] \\ \text{(B)} & [\text{MLT}^{-2} \text{ A}^{-2}] \\ \text{(C)} & [\text{ML}^{-2} \text{ T}^{-2} \text{ A}^{-1}] \\ \text{(D)} & [\text{MLT}^{-2} \text{ A}^{-1}] \end{array}$$

5. Two parallel straight wires carrying currents I_1 and I_2 are separated by a distance x . The current in the first wire is increased from I_1 to $2I_1$ and their separation is reduced to $\frac{x}{2}$. The value of force between the wires will :
- (A) become four times
 - (B) become two times
 - (C) becomes one-fourth
 - (D) remain the same
6. A bar magnet is thrown from a distance into a closed loop with its north pole towards the loop and passes through it. The induced current in the loop is :
- (A) first clockwise and then anticlockwise
 - (B) first anticlockwise and then clockwise
 - (C) clockwise only
 - (D) anticlockwise only
7. The speed of electromagnetic waves in a medium depends upon :
- (A) properties of the medium
 - (B) frequency of the waves
 - (C) wavelength of the waves
 - (D) intensity of the waves
8. The phase difference between any two points on a wavefront of a wave is :
- | | |
|-----------|---------------------|
| (A) π | (B) $\frac{\pi}{2}$ |
| (C) Zero | (D) $\frac{\pi}{6}$ |

9. The ratio of the de Broglie wavelengths associated with an electron of mass m and of proton of mass M when both are accelerated through a potential difference V is :
- (A) $\left(\frac{M}{m}\right)^{3/2}$ (B) $\frac{m}{M}$
- (C) $\sqrt{\frac{M}{m}}$ (D) $\sqrt{\frac{m}{M}}$
10. The electron of a hydrogen atom is in the second excited level. During its de-excitation to the ground state, the number of possible spectral lines that can be emitted is :
- (A) 1 (B) 2
- (C) 3 (D) 6
11. The majority and the minority charge carriers inside a p-type semiconductor are respectively :
- (A) electrons, holes
- (B) holes, electrons
- (C) holes, holes
- (D) electrons, electrons
12. The output of a full-wave rectifier is :
- (A) steady
- (B) continuous with ripple
- (C) continuous without any ripple
- (D) zero during alternate half cycles of input

Questions number 13 to 16 are Assertion (A) and Reason (R) type questions. Two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (A), (B), (C) and (D) as given below.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
- (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).
- (C) Assertion (A) is true, but Reason (R) is false.
- (D) Assertion (A) is false and Reason (R) is also false.

13. Assertion (A) : The kinetic energy of the emitted photoelectrons depends upon the work function of the surface.

Reason (R) : The maximum value of kinetic energy of photoelectrons is the difference of energy of incident photons and work function of the surface.

14. Assertion (A) : At a very high frequency, a capacitor can behave as a conductor.

Reason (R) : The capacitance of a capacitor depends only on its geometrical configuration and nature of dielectric used.

15. Assertion (A) : A magnetic field does not interact with a moving charge.

Reason (R) : A moving charge does not produce a magnetic field.

16. Assertion (A) : In optical fibres, the refractive index of cladding must be lesser than that of the core.

Reason (R) : Total internal reflection takes place only when the ray passes from denser to rarer medium.

SECTION B

17. Two conducting wires of same material have radii r and $\frac{r}{2}$. The current flowing through them is I and $2I$, respectively. Find the ratio of drift velocity of free electrons in the wires. 2
18. (a) Why does a water tank appear shallow ? Explain. Write the relation between real depth and apparent depth of the tank with the refractive index of water. 2
- OR**
- (b) What are optical fibres ? How does a light ray travel through an optical fibre ? Give two applications of optical fibres. 2
19. A monochromatic light beam is incident on a slit of width 0.2 mm. The diffraction pattern is formed on a screen placed 1.5 m away from the slits. The first maximum is formed at a distance of 3.0 mm from the centre of the screen. Calculate the wavelength of light used. 2
20. What are the postulates of Bohr's theory of hydrogen atom ? Write the expression of radius of n^{th} orbit of electron. 2
21. Explain the formation of potential barrier across an unbiased p-n junction. Write the effect on height of the barrier when a p-n junction is (a) forward biased, and (b) reverse biased. 2

SECTION C

22. (a) Define mobility of free electrons in a conductor. How does the mobility of electrons in a metal change when :

(i) temperature of the conductor is decreased at constant potential difference ?

(ii) applied potential difference is doubled at a constant temperature ?

3

OR

(b) Obtain an expression for emf and internal resistance of a cell which is equivalent to parallel combination of two cells of emf and internal resistance (E_1, r_1) and (E_2, r_2), respectively.

3

23. (a) Define coefficient of self-inductance.

(b) A solenoid has length 12 cm, area of cross-section 24 cm^2 and 1500 turns. Calculate the self-inductance of the solenoid.

3

24. A coil of area of cross-section 'A' has N number of turns. It is rotating with an angular speed ω inside a uniform magnetic field B. If the total resistance of circuit of coil is R, find :

3

(a) the maximum current in the circuit.

(b) the flux through the coil when the current is zero.

(c) the flux through the coil when the current is maximum.

25. Explain the polarization of a dielectric medium filled inside a charged capacitor. How does the capacity of a capacitor increase due to filling ? What is dielectric constant of a dielectric ? 3
26. A glass biconvex lens of focal length 0.6 m and refractive index 1.5 is immersed in a medium of refractive index 2.0. Find the new power of the lens and the nature of lens. 3
27. The threshold frequency for a metal surface for photoelectric emission is 4×10^{14} Hz. When a light of frequency ν is incident upon it, the emitted photoelectrons are stopped by a stopping potential of 4.98 V. Calculate : 3
- (a) work function of the metal, and
- (b) the value of ν .
28. (a) Explain the formation of valence and conduction band in a crystal.
- (b) Discuss the difference between energy bands of (i) a metal, (ii) an insulator, and (iii) a semiconductor. 3

SECTION D

Questions number 29 and 30 are case study based questions. Read the following paragraphs and answer the questions that follow.

29. The electric potential at a point in an electric field is the work done in bringing a unit positive charge from infinity to this point. If the potential difference between any two points at a surface is zero, the surface is called an equipotential surface. The shape of an equipotential surface can give direction of electric field at a point on it. It may be a closed surface or an open surface, depending on the charges creating the electric field.
- (i) The shape of equipotential surface due to an isolated point charge is : 1
- (A) ellipsoidal, with charge at its one foci
- (B) plane surface not passing through point charge
- (C) spherical with charge at its centre
- (D) cylindrical with charge at its axis

- (ii) The equipotential surfaces in a uniform electric field acting along + Z direction are : 1
- (A) planes parallel to the XY plane
 - (B) concentric spherical surfaces
 - (C) planes parallel to the YZ plane
 - (D) planes parallel to the XZ plane
- (iii) Which of the following statements is **not** true for equipotential surfaces ? 1
- (A) Potentials at two equipotential surfaces are different.
 - (B) No work is done in moving a charge on an equipotential surface.
 - (C) Equipotential surfaces always enclose some volume.
 - (D) Two equipotential surfaces do not intersect each other.
- (iv) (a) The angle between the electric field at a point and outward normal at that point on an equipotential surface is : 1
- (A) 90° (B) 45°
 - (C) 30° (D) 0°

OR

- (iv) (b) A and B are two equipotential surfaces around a point charge Q. Surface A is closer to the charge than surface B. If a test charge is released between them, it will : 1
- (A) remain stationary
 - (B) move from A to B
 - (C) move from B to A
 - (D) move around Q in a circular path between the equipotential surfaces A and B

30. When an electron of hydrogen atom jumps from a ground state energy level to higher energy level, it absorbs energy and is excited. It can remain in this state for a maximum time $\simeq 10^{-7}$ s. When de-excited, the electron returns back and the same amount of energy is radiated in the form of a photon. The difference of energies of two energy levels ($E_f - E_i$) is the energy of this photon. Depending upon the two levels involved in transition, the spectral lines are grouped in five series of hydrogen line emission spectrum. These are named as Lyman ($n_f = 1$), Balmer ($n_f = 2$), Paschen ($n_f = 3$), Brackett ($n_f = 4$) and Pfund ($n_f = 5$) series. The values of energy of the electron in the n^{th} level of hydrogen atom and of hydrogen-like atom are $\frac{-13.6}{n^2}$ eV and $\frac{-13.6}{n^2} Z^2$ eV respectively. Here Z is number of protons in the nucleus of hydrogen-like atom.

(i) Which one of the following hydrogen spectral series lie in visible region ? 1

- | | |
|-----------|-------------|
| (A) Lyman | (B) Balmer |
| (C) Pfund | (D) Paschen |

(ii) In a hydrogen atom, when an electron jumps from $n = 6$ to $n = 3$ level, the spectral line obtained belongs to : 1

- | | |
|--------------|-------------|
| (A) Lyman | (B) Pfund |
| (C) Brackett | (D) Paschen |

(iii) The second member of Lyman spectral series in absorption hydrogen spectrum results due to electron transition : 1

- | | |
|------------------------|------------------------|
| (A) $n = 3$ to $n = 1$ | (B) $n = 1$ to $n = 3$ |
| (C) $n = 1$ to $n = 2$ | (D) $n = 2$ to $n = 1$ |

- (iv) (a) The ratio of energies in Bohr hydrogen atom and in hydrogen-like He atom for the first electron orbit is : 1
- (A) 1 : 4 (B) 4 : 1
(C) 2 : 1 (D) 1 : 2

OR

- (iv) (b) The spacing between successive energy levels of hydrogen atom : 1
- (A) is same for all levels
(B) becomes larger for higher levels
(C) becomes smaller for higher levels
(D) may be larger or smaller for higher levels depending upon temperature

SECTION E

31. (a) (i) Define electric field. Write its SI units. Derive an expression for electric field due to a point charge at a point. Why must the test charge be as small as possible ?
- (ii) An electron and an α -particle, both are kept in the same electric field $\vec{E} = E_1 \hat{i} + E_2 \hat{j}$. Find the force acting on them and their acceleration. 5

OR

- (b) (i) Obtain an expression for the capacitance of a parallel plate capacitor.
- (ii) Two identical plates, each of area A having surface charge densities $+\sigma$ and $-\sigma$, respectively, are separated by a distance d in air and a dielectric of dielectric constant K is filled between them. Write the expressions for : 5
- (1) the potential difference between the plates
 - (2) capacitance of the capacitor so formed
 - (3) energy stored in this capacitor

32. (a) (i) Discuss the motion of a charged particle of mass m and charge q in a uniform magnetic field $\vec{B}$ with initial velocity $\vec{v}$
- (1) at an angle θ with field direction,
 - (2) perpendicular to the field direction ($\theta = 90^\circ$), and
 - (3) parallel to the field direction ($\theta = 0^\circ$).
- (ii) An electron is moving along a circular path of radius 18.2×10^{-2} m in a uniform magnetic field 2×10^{-3} T. Calculate :
- (1) the speed of the electron, and
 - (2) the potential difference through which the electron must be accelerated to acquire this speed.

5

OR

- (b) (i) Deduce an expression for the torque experienced by a current carrying coil placed in an external uniform magnetic field.
- (ii) The maximum torque acting on a current carrying loop of area 0.08 m^2 is 8×10^{-8} N-m. The current in the loop is $1.6 \mu\text{A}$. Calculate :
- (1) the magnetic field in which the loop is kept, and
 - (2) the magnetic dipole moment of the loop

5

33. (a) (i) Write the relation between u , v and R for a convex spherical surface when refraction takes place from optically rarer to denser medium. Using it, derive 'lens maker formula' for a double convex lens.
- (ii) Two thin lenses of power $+ 25 \text{ D}$ and $- 5 \text{ D}$ are kept in contact with each other.
- (1) Calculate the focal length of the combination.
 - (2) An object of 4 cm height is placed at a distance of 25 cm from this combination. What is the position and size of the image formed ?

5

OR

- (b) (i) Give reasons for the following :
- (1) Why is an objective lens of large focal length and large aperture used in a telescope ?
 - (2) Two sodium lamps (independent) cannot emit coherent lights.
 - (3) The resultant intensity at any point on the screen varies between zero and four times the intensity, due to one slit, in Young's double slit experiment.
- (ii) In Young's double slit experiment, fringes are obtained on a screen using two slits separated by 2 mm. When the screen is moved 20 cm towards the slit, the change in fringe width is found to be 6×10^{-5} m. Calculate the wavelength of the light used.

5